INCH-POUND

MIL-C-24643/53D
22 November 1994
SUPERSEDING
MIL-C-24643/53C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 3000 VOLTS, TYPES LSSRW, LSDRW AND LSTRW
* (INCLUDING VARIATIONS TYPES LSSRWA, LSDRWA AND LSTRWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: TYPES LSSRW, LSDRW AND LSTRW (unarmored) and LSSRWA, LSDRWA AND LSTRWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, AWG size 14, class B stranding.
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Insulation wall thickness 0.080 inch minimum average. Standard identification code shall be applied by method 1 or method 3. The insulation shall be colored black if method 1 is used.
Fourth - The required two or three conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap for two and three conductor cables only.

Construction variant for type LSSRW, LSDRW AND LSTRW

Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.


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Construction (watertight) (continued)

Construction variant for type LSSRWA, LSDRWA AND LSTRWA

Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.
Seventh- Braided metal armor.

TABLE I. Details

| | Military[1/] part no. M24643/53 | Cable type | Number of conductors | Overall diameter maximum (inch) | Conductor resistance (ohms) | Cold bending mandrel diameter (max inch) |
|---|---|---|---|---|---|---|
| | -01UN | LSSRW | 1 | 0.400 | 2.69 | 3 |
| | -02UN | LSDRW | 2 | .670 | 2.75 | 5 |
| | -03UN | LSTRW | 3 | .710 | 2.75 | 7 |
| * | -04AN | LSSRWA | 1 | .450 | 2.69 | 3 |
| * | -05AN | LSDRWA | 2 | .720 | 2.75 | 5 |
| * | -06AN | LSTRWA | 3 | .760 | 2.75 | 7 |

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.

EXAMINATION AND TESTS:

| Basic electrical: | Requirements |
|---|---|
| Conductor Resistance - ohms/1000 feet at 25°C, maximum | (see table I) |
| Voltage withstand - volts, root mean square, minimum | 8000 |
| Insulation resistance - Conductor to conductor, Megohms/1000 feet, minimum | 9500 |
| Conductor continuity | No failure |

| Group A: | |
|---|---|
| Visual and dimensional | No failure |
| Watertightness - see MIL-C-24643 for water leakage | No failure |

| Group B: | |
|---|---|
| Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200°C) | 50 |
| Cold bending, cable - (see table I for mandrel diameter) | No damage |
| Drip - 95 ±1°C | Zero |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength - $lb/in^2$, minimum | 1800 |
| Elongation - percent, minimum | 250 |



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EXAMINATION AND TESTS (continued):

| | Requirements |
|---|---|
| Jacket (cable) | |
| Tensile strength - lb/in$^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Group C: | |
| Physicals - (aged) | |
| Insulation | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged minimum | 80 |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged minimum | 60 |
| Hot oil immersion | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 30 |
| *Permanence of printing* (conductor) - (method 1 only) cycles, minimum (LSDRW and LSTRW) | 25 |
| * Permanence of printing (jacket) - cycles, minimum | 125 |
| Cable sealant removability | No failure |
| Armor - conformance to material, construction and coverage (type LSSRWA, LSDRWA and LSTRWA only) | No failure |
| Group D: | |
| Flame propagation (cable) | No failure |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

| | Requirements |
|---|---|
| Aging and compatibility (cable) (125 ±5°C.) | No failure |
| Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |



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QUALIFICATION INSPECTION (continued):

| | Requirements |
|---|---|
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) |
|---|---|
| LSSRW | 2500 |
| LSDRW and LSTRW | 2000 |

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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